1(a)	(gravitational) force is (directly) proportional to product of masses	**B1**
	force (between point masses) is inversely proportional to the square of their separation	**B1**
1(b)	correct read offs from the graph with correct power of ten for R^3	**C1**
	$M = \dfrac{4 \times \pi^2 \times 1.2 \times 10^{34}}{6.67 \times 10^{-11} \times 2.4 \times (365 \times 24 \times 3600)^2}$	**C1**
	$= 3.0 \times 10^{30}$ kg	**A1**
1(c)(i)	potential energy is zero at infinity	**B1**
	(gravitational) forces are attractive	**B1**
	work must be done on the rock to move it to infinity	**B1**
1(c)(ii)	$\dfrac{GMm}{r^2} = \dfrac{mv^2}{r}$ *OR* $v^2 = \dfrac{GM}{r}$ *OR* $v = \sqrt{\dfrac{GM}{r}}$	**M1**
	use of ½ mv^2 (e.g. multiplication by ½ m) leading to $\dfrac{GMm}{2r}$	**A1**
1(c)(iii)	Ep = ϕm <u>and</u> $\phi = \dfrac{-GM}{r}$ or $E_p = \dfrac{-GMm}{r}$ Total energy = $E_k + E_p$	**C1**
	Total energy $= \dfrac{GMm}{2r} + \dfrac{-GMm}{r} = \dfrac{-GMm}{2r}$	**A1**